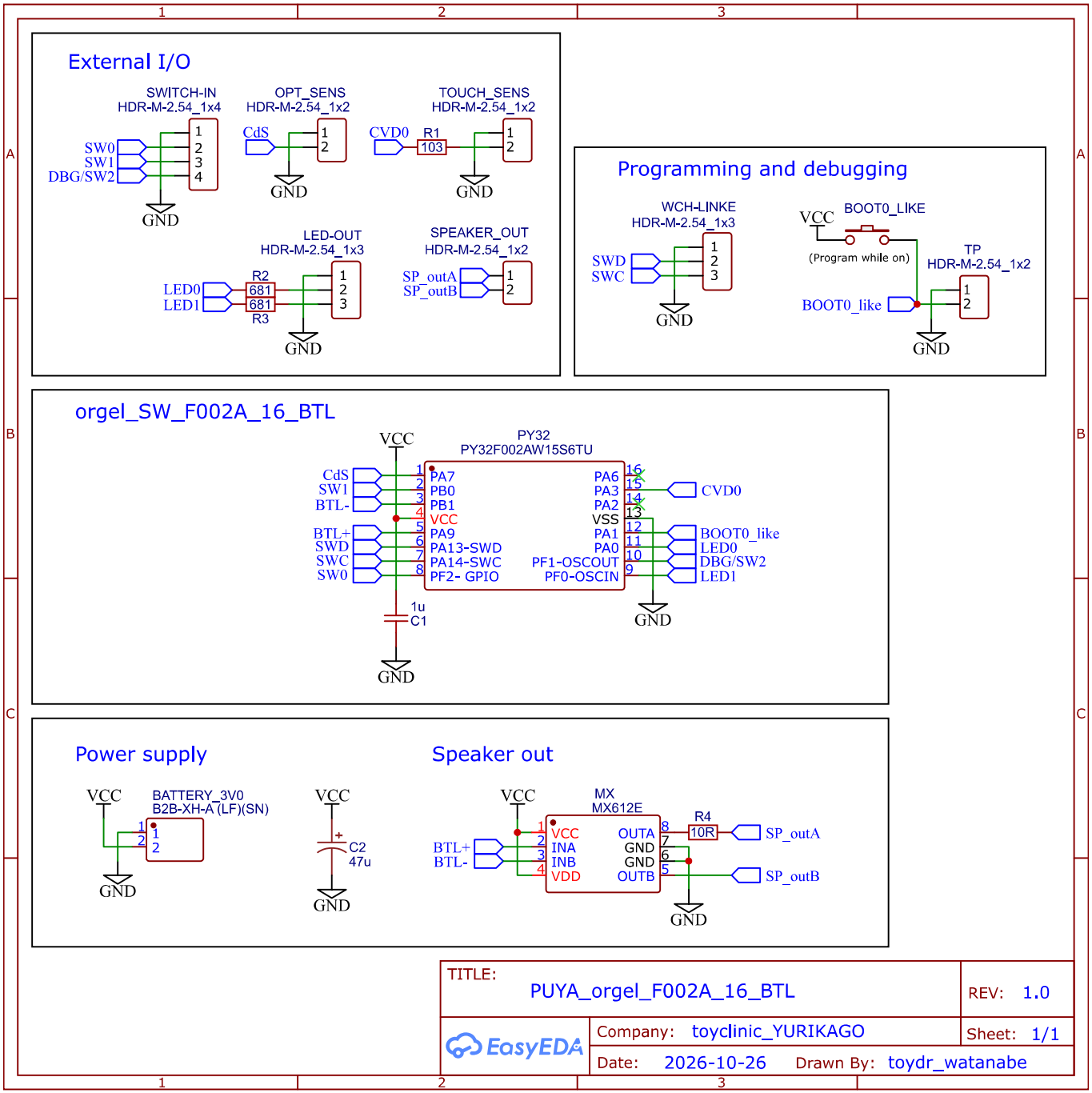

External I/O
SWITCH-IN
HDR-M-2.54_1x4
SW0
SW1
DBG/SW2
GND
OPT_SENS
HDR-M-2.54_1x2
CdS
GND
TOUCH_SENS
HDR-M-2.54_1x2
CVD0
R1
103
GND
LED-OUT
HDR-M-2.54_1x3
LED0
LED1
R2
681
681
R3
GND
SPEAKER_OUT
HDR-M-2.54_1x2
SP_outA
SP_outB
Programming and debugging
WCH-LINKE
HDR-M-2.54_1x3
SWD
SWC
GND
VCC
BOOT0_LIKE
(Program while on)
TP
HDR-M-2.54_1x2
BOOT0_like
GND
orgel_SW_F002A_16_BTL
PY32
PY32F002AW15S6TU
VCC
CdS
SW1
BTL-
BTL+
SWD
SWC
SW0
PA7
PB0
PB1
VCC
PA9
PA13-SWD
PA14-SWC
PF2- GPIO
PA6
PA3
PA2
VSS
PA1
PA0
PF1-OSCOUT
PF0-OSCIN
CVD0
BOOT0_like
LED0
DBG/SW2
LED1
GND
1u
C1
GND
Power supply
VCC
BATTERY_3V0
B2B-XH-A(LF)(SN)
GND
VCC
C2
47u
GND
Speaker out
VCC
MX
MX612E
BTL+
BTL-
VCC
INA
INB
VDD
OUTA
GND
GND
OUTB
R4
10R
SP_outA
SP_outB
GND
TITLE:
PUYA_orgel_F002A_16_BTL
REV: 1.0
EasyEDA
Company: toyclinic_YURIKAGO
Sheet: 1/1
Date: 2026-10-26
Drawn By: toydr_watanabe